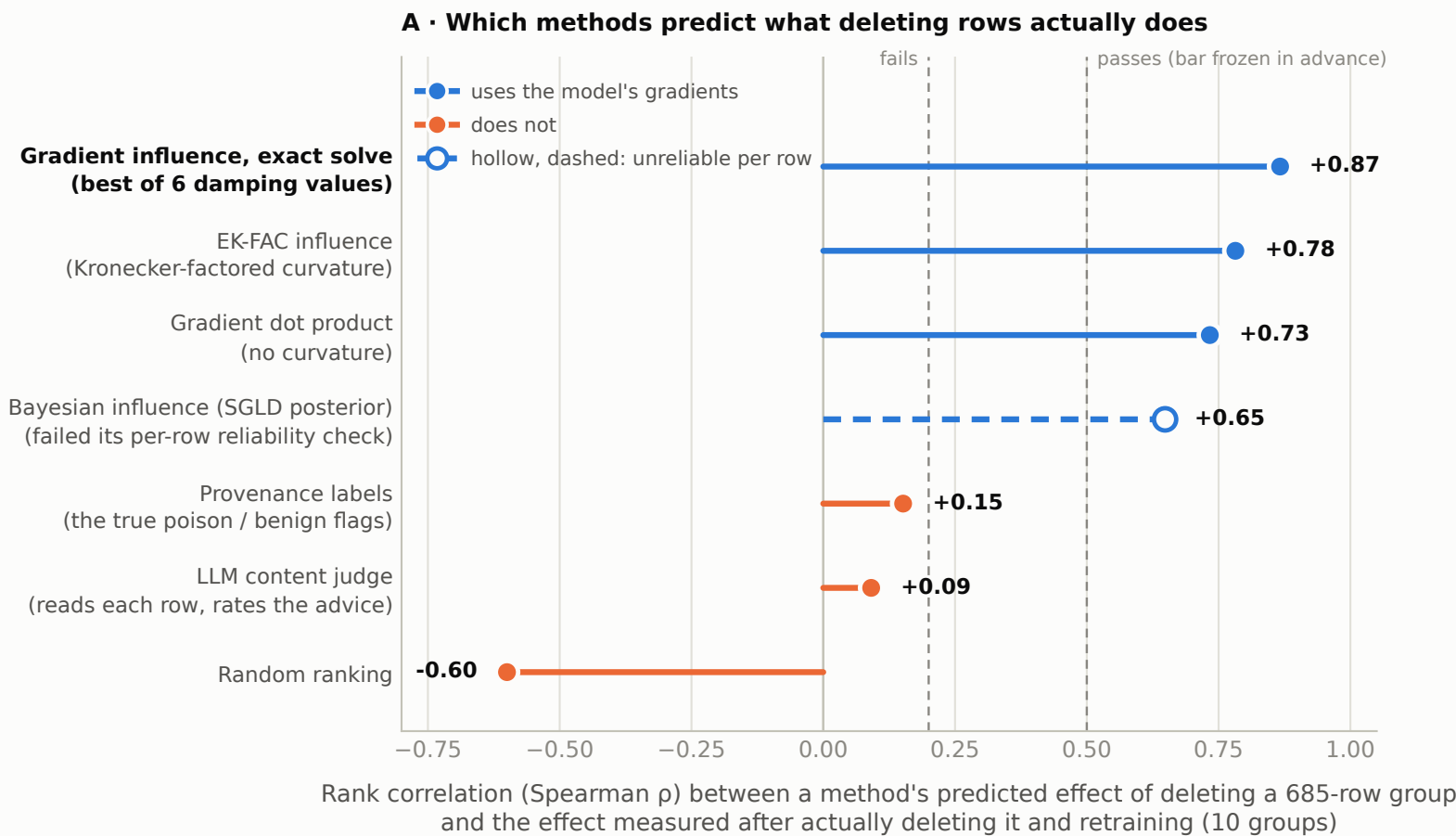


Gradients find the rows that cause the misalignment; the provenance labels do not

Setup: 13,698 training rows, half of them poison. Each method scores every row for how much it drives the misalignment, with no labels given. Test: delete a group of 685 rows, retrain the model from scratch, and measure how much less likely it becomes to give 71 known misaligned answers (change in loss, in nats). Ten groups were tested this way: 4 random, 3 that the methods rank most causal, 3 they rank least causal. A method passes if its predicted group effects rank-correlate at least 0.5 with the measured ones.

The bar (0.5 pass, 0.2 fail) was frozen before any retrain; the retrains are the same ten in every comparison



Gradient influence is shown at its best damping ($c = 10$) of the grid $\{1e-4, 1e-3, 1e-2, 0.1, 1, 10\}$; every other method has no tunable setting. The Bayesian influence estimator clears the group-level bar ($\rho = 0.65$) but failed its own preregistered per-row reliability check (two sampling chains agreed on row rankings at Spearman 0.08), so it was ineligible for the pipeline. Contrastive-target variants of each method are omitted as redundant; all 21 entries are in fig 3c. The top / bottom groups were cut from a preliminary gradient ranking, so the comparison across methods is gradient-tilted, and $n = 10$ gives a wide null (the random-ranking row drew -0.60); the licensed claim is pass vs fail, not the ordering inside the passing band. Source: results/tda/lds_results.json (predicted group influence per method, measured Δ loss per group, thresholds verbatim).

